GRAYCO
CONSTRUCTION
SERVICES

GRAYCO STUDY GUIDE

Odor Control
Key Terms

Every key term from the course with a plain-English definition, grouped by subject. Use it to review before the GrayCo test and the IICRC OCT exam.

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KEY TERMS	SUBJECTS

GRAYCO CONSTRUCTION SERVICES, LLC · BARTLETT, TENNESSEE

STUDY GUIDE

Key terms and definitions

01 Introduction

The nose
Odors are gases that come from a source into the air and are perceived by the nose. The technician's best instrument for detecting the presence and intensity of odor.

Olfactory lobe
The part of the brain where humans recognize and identify odors.

Real odor
The actual sensation of smell: odor molecules contact olfactory nerve endings in the nose and the signal goes to the olfactory lobe.

Heightened awareness odor / psychological odor
Odors people believe they smell, based on experiences, suggestions and past impressions.

Water soluble
To be smelled, an odor molecule must be at least slightly soluble in water, or it can't get through the watery film on the nerve endings. (It must also be volatile and soluble in lipids.)

Acclimated
People can become accustomed to an odor and stop noticing it (the baker doesn't smell the bread).

Temperature and relative humidity
Odor is most apparent when both are high.

Four principles of odor control
1) Remove as much of the odor source as possible. 2) Clean the contaminated (source) area. 3) Penetration of odor counteractant into all surfaces and spaces. 4) Sealing.

Sealers
Lock in or prevent further release of a malodor. Rated by perm (permeance) factor; the closer to zero, the more effective.

02 Odor Removal Procedures

Adsorbents
Solids like activated charcoal or baking soda that capture and hold odors on their surfaces.

Neutralization
A chemical balance (1 – 1 = 0); works on chemical odors like chlorine and acids. Baking soda neutralizes acid spill odor.

Oxidation
A chemical reaction in which oxygen combines with another substance. Oxidizing agents: hydrogen peroxide, chlorine bleach, ozone (from an ozone machine).

Biocides
Materials that inhibit or kill microorganisms. Disinfectants and antimicrobials are biocides used for microbial odors (MVOCs).

Counteraction
To "work against." Complex odors like smoke respond to suppression spraying, vapor phase re-odorization and thermal fogging.

Masking and pairing agents
Most smoke odor products contain both. Pairing agents combine with and change malodors into odorless compounds.

Masking
Disguising a malodor with a stronger, more pleasant odor. Time-release deodorants (granules, oils, gels) are masking agents.

Bioenzymatic digestion
Engineered bacteria produce enzymes that digest odorous material such as animal waste, spilled fuel oil and decaying protein (milk, eggs, fish).

Micron
Unit for droplet/particle size: one thousandth of a millimeter, or 1/25,000 of an inch.

Pressure / compression sprayers
Droplets often hundreds of microns. Most control over where product goes; good for coating surfaces.

Wet foggers / misters
Mechanical action with air pressure and nozzles; droplets 25–100 microns. Inexperienced use can over-wet and stain moisture-sensitive surfaces.

ULV (ultra-low volume) foggers
Cold-fog water- or alcohol-based formulations; droplets 8–15 microns that stay airborne 5–6 hours.

Thermal foggers
Use heat to make 0.5–2 micron droplets of petroleum-based deodorant; modify and counteract malodors.

Ozone (O3)
Oxidizing gas very effective on organic odors (skunk, trauma, dead animals, natural-fuel fires). Made by an ozone generator. Heavier than air, so circulate with a fan. Corrosive; harms plants and natural rubber.

Ozone exposure
Low levels: sharp, irritating odor. Higher: eye irritation, dry nose (nosebleeds) and throat, sneezing, coughing.

Ozone generator
When turned off, most ozone dissipates in 10–15 minutes. Ventilate thoroughly before re-occupancy.

Exhaust fans and air scrubbers
Remove or dilute malodors and particulates; exhaust treated spaces for faster re-occupancy.

03 Removing Smoke Odors

Protein odors
Persistent malodors from burned meat or poultry. Yellow/brown, greasy residue; wet cleaning often required.

Natural substance odors
From burned wood, paper, cotton, wool, etc. Gray or black, dry and powdery.

Synthetic substance odors
From burned plastics and synthetic textiles. Black residue that smudges easily.

Heat
Heated air rises and migrates to cold areas; heat opens pores.

Pressure
Fire energy drives smoke into minute cracks and crevices.

Impingement
Splatter; a substance hits a surface fast enough to stick.

Magnetism
Smoke is attracted to metal surfaces (pipes, nail heads, hangers).

Ionization
Opposite charges attract, forming smoke webs.

Driven smoke
Hot smoke, pressurized; found on vertical surfaces.

Free floating smoke
Cold smoke that has lost its energy; heavier than air, settles on horizontal surfaces (where most smoke is deposited).

Increases
Higher temperature is the biggest single factor that increases odor penetration and intensity.

Multiple techniques
The best system for smoke odor (the three-step Unsmoke System). One method alone gives incomplete results.

Precede
Smoke odor removal should precede normal cleaning and any repairs.

Step 1 – Suppression spray
Spray all surfaces in the source area (preliminary detoxification) with water-based odor counteractant.

Step 2 – Time-released odor counteractant
Granules applied to horizontal surfaces, attics, crawl spaces, plenums.

Step 3 – Thermal fogging
Duplicates the heat and penetration of the fire.

Thermal fogging safety
Evacuate unprotected persons, pets and houseplants; wear respiratory, eye and skin protection; inform the fire department.

Possible explosion
The most serious hazard of thermal fogging petroleum products in a confined area; turn off pilot lights.

Oily
Over-applying oily fogging products creates slip hazards and oily residue.

04 Controlling Microbial Odors

Pathogenic
Harmful, disease-causing microorganisms.

Bacteria
One-celled microorganisms; reproduce by fission, as often as every 15–30 minutes.

Fungi
Large group of plant-like organisms that feed on living or dead organisms; lack chlorophyll.

68°–86°F (20°–30°C)
Temperature range where fungi grow, the range people find comfortable.

Fungal odors / MVOCs
Odors from the microorganism's digestion, called microbial volatile organic compounds.

Germination / spores
Moisture is essential for germination and growth of fungi; many fungi start as spores.

Organic materials
Fungi from water damage prefer cellulose in natural fibers and wood.

Viruses
Parasites that only live and reproduce inside cells of a living host.

EPA / Health Canada
Agencies that register and regulate disinfectants and antimicrobials (U.S. and Canada).

Antimicrobials
Biocides for microbial and moisture-related odors; keep the surface wet for the label's dwell time.

Label directions
Following product label directions is critical; the label legally includes flyers and technical bulletins.

Sterilant
Kills microorganisms and bacterial spores.

Disinfectant
Kills microorganisms, but not necessarily the spores.

Sanitizer
Reduces microorganisms to a level safe for human use.

-cide / -stat
-cide = lethal to organisms. -stat = limits or controls the number of organisms.

Danger / Warning / Caution
Label signal words: highly toxic / moderately toxic / low toxicity.

05 Controlling Animal Odors

Guarantee
Never guarantee urine odor removal if the pet stays in the structure.

Discarding
Discard contaminated carpet cushion (padding).

Bio-enzymatic
Bio-enzymatic digesters may be needed for complete removal of severe urine odor.

Penetration
Know the degree of penetration into flooring and building materials.

Multiple layers
Gravity pulls urine into the layers of carpet, pad and subfloor, making it hard to treat.

Black light / ultraviolet (UVA) light
The most reliable device to locate urine contamination.

High intensity lamp / moisture sensors
Other tools to find urine not visible in normal light.

Acidic / alkaline salts
Fresh urine is acidic; dried urine is alkaline salts.

Urine salts
Hygroscopic (moisture loving); attract moisture and release odor. Not readily water soluble; removed with acid rinse.

Customer-applied / spot treatments / total urine decontamination
Treatment options from least to most expensive and effective. Spot treatment: acidic rinse, flushing with clear water, bio-enzymatic, dwell time, thorough extraction.

Concrete surfaces
Easiest to decontaminate; acid cleaner etches concrete and neutralizes alkaline salts.

Vinyl tile and linoleum
Most difficult to decontaminate; usually removed.

Neutralize the alkaline salt
Apply acid solution, agitate, allow 20–30 minutes dwell.

Nonporous sealing compound
Seal odor-damaged surfaces before painting (e.g., pigmented alcohol/shellac sealer, lacquer, alkyd paint).

Tack strip
All contaminated tack strip is removed and discarded.

Carcass
Locating it is the biggest obstacle with dead animals (borescope, cadaver dogs, wetting and heating).

Chlorine bleach (sodium hypochlorite)
Inexpensive liquid oxidizer for skunk odor on durable colorfast surfaces. Can discolor nylon/fabrics, corrode metal, destroy protein fibers.

Oxidize the source
Skunk and decomposition odors are removed by oxidation reactions.

Hydrogen peroxide
Neutralizes skunk odor on colorfast surfaces; may remove color from fabric.

06 Controlling Trauma Scene Odors

Hepatitis B and HIV
The two bloodborne pathogens of concern. Hepatitis B affects the liver; HIV leads to AIDS.

Proteinaceous materials
Decomposition of protein causes death scene odor.

Bloodborne pathogen
A microorganism in blood, body discharge or other potentially infectious materials (OPIM).

Exposure control plan
First step of a bloodborne pathogen response program; outlines how employees could be exposed and how to control it.

Mandatory PPE training
Wearing PPE is mandatory whenever there is risk of exposure to biohazards.

Universal precautions
Treating all blood, body fluids and contaminated materials as potentially infectious.

Biohazardous material
Use bio-disposal bags; labels must be on fluorescent red-orange paper.

HBV immunization
Employers must offer it free; three injections. If exposed and not immunized, first shot within 24 hours.

Exposure incidents
Eye splash: flush 15 minutes with clean running water. Skin: wash with soap and water. Employer must document the route and circumstances.

Decontamination / disinfection
Rendering bloodborne pathogens incapable of transmitting infection so the item is safe to handle, use or dispose of.

Disinfectant (final step)
All salvageable surfaces touched by blood/OPIM are cleaned with an appropriate disinfectant, then porous surfaces are sealed.

07 Structure and Building Systems

Porous surfaces
ETS odor comes mainly from smoke residue accumulated on surfaces, especially porous ones.

0.01 to 1.0 microns
Size range of ETS (environmental tobacco smoke) particles.

Washing affected surfaces
The critical component of ETS deodorization.

Three-step process
Step 1 suppression spray, Step 2 odor counteractant and absorbent, Step 3 thermal fogging.

Never fog forced air systems
Never fog or spray liquid deodorizers into forced air systems.

The source
The biggest mistake in odor removal is failing to locate and remove the source.

08 Contents and Furnishings

Ventilation
The recommended first step for new carpet odor (4-PCH from latex adhesive), sometimes with bake-out and adsorption.